

Lectures 1 and 2: Gaussians

Definition, linear structure, and conditioning as projection

OUTLINE $\varphi_X = \exp(\text{quadratic}) \rightarrow AX + b$ is Gaussian $\rightarrow$ uncorrelated $\Rightarrow$ independent $\rightarrow$ conditioning is orthogonal projection $\rightarrow$ ridge regression, and denoising

HOW TO READ Sections marked * are not lectured. They are complete, they are examinable, and they carry most of the exercises. Exercises are typed A (did you follow the lecture), B (did you follow the argument) and C (one idea, short once you have it). Each week's quiz has one of each.

1 Gaussians, and their linear structure

1.1 Some facts from Fourier analysis

The following is standard, and it is the only material in the course we do not prove. Fix the convention now, since every sign below depends on it: the transform carries $e^{-ix \cdot \xi}$ and the inverse carries $e^{+ix \cdot \xi}$.

Proposition 1.1 (Fourier analysis, imported). *For $f \in L^1(\mathbb{R}^d)$ and a finite measure ν on $\mathbb{R}^d$ define*

$$\widehat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-ix \cdot \xi} dx, \quad \widehat{\nu}(\xi) = \int_{\mathbb{R}^d} e^{-ix \cdot \xi} d\nu(x), \quad \xi \in \mathbb{R}^d.$$

Then:

- (i) **Inversion.** *If f and $\widehat{f}$ both lie in $L^1(\mathbb{R}^d)$ then $f(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} \widehat{f}(\xi) e^{ix \cdot \xi} d\xi$ at every point of continuity of f .*
- (ii) **Derivatives.** *If $f, \partial_j f \in L^1$ then $\widehat{\partial_j f}(\xi) = i\xi_j \widehat{f}(\xi)$; if $x_j f \in L^1$ then $\partial_{\xi_j} \widehat{f}(\xi) = \widehat{(-ix_j f)}(\xi)$. Differentiating in x is multiplication by $i\xi_j$, and multiplication by x_j is $i\partial_{\xi_j}$.*
- (iii) **Isometry.** *For $f \in L^1 \cap L^2$ one has $\|\widehat{f}\|_{L^2}^2 = (2\pi)^d \|f\|_{L^2}^2$, and the transform extends to a bijection of $L^2(\mathbb{R}^d)$ onto itself.*
- (iv) **Uniqueness.** *If μ and ν are finite complex measures on $\mathbb{R}^d$ with $\widehat{\mu} = \widehat{\nu}$, then $\mu = \nu$.*

Part (iv) is what lets a law be *defined* by its transform, which is what we do next, and it is used a second time before this lecture ends. Part (ii) is the engine of almost every Gaussian computation in the course; part (i) is used once, in Lectures 3–4.

1.2 The definition

Definition 1.2 (Characteristic function). The *characteristic function* of a random vector X in $\mathbb{R}^d$ with law ν is the Fourier transform of ν :

$$\varphi_X(\xi) := \widehat{\nu}(\xi) = \mathbb{E}[e^{-iX \cdot \xi}], \quad \xi \in \mathbb{R}^d.$$

It is defined for every X , with no integrability assumption, because $|e^{i\theta}| = 1$. Immediately $\varphi_X(0) = 1$, $|\varphi_X(\xi)| \leq 1$ and $\varphi_X(-\xi) = \overline{\varphi_X(\xi)}$.

Definition 1.3 (Gaussian vector). Let $\mu \in \mathbb{R}^d$ and let Σ be a symmetric $d \times d$ matrix. A random vector X in $\mathbb{R}^d$ is *Gaussian with mean μ and covariance Σ* , written $X \sim \mathcal{N}(\mu, \Sigma)$, if

$$\varphi_X(\xi) = \exp\left(-i\mu \cdot \xi - \frac{1}{2}\xi^\top \Sigma \xi\right) \quad \text{for every } \xi \in \mathbb{R}^d.$$

By Proposition 1.1(iv) at most one law satisfies this. Whether any does is the business of §1.3, and the answer will be: exactly when $\Sigma \in \text{Sym}_{\geq 0}(\mathbb{R}^d)$, so nothing is lost by reading the definition with that restriction imposed from the start. First, the two words in the definition have to be earned.

Proposition 1.4 (The parameters are what they are called). *If $X \sim \mathcal{N}(\mu, \Sigma)$ then $\mathbb{E}|X|^2 < \infty$, and*

$$\mathbb{E}[X_j] = \mu_j, \quad \mathbb{E}[(X_j - \mu_j)(X_k - \mu_k)] = \Sigma_{jk}.$$

In particular the law determines (μ, Σ) , so two Gaussians agree in law if and only if their parameters agree.

Proof. Write $\psi(\xi) = -i\mu \cdot \xi - \frac{1}{2}\xi^\top \Sigma \xi$, so $\varphi_X = e^\psi$ is smooth and

$$\partial_j \varphi_X|_{\xi=0} = -i\mu_j, \quad \partial_j \partial_k \varphi_X|_{\xi=0} = (\partial_j \partial_k \psi + \partial_j \psi \partial_k \psi)|_{\xi=0} = -\Sigma_{jk} - \mu_j \mu_k.$$

Finiteness of the second moment is not an assumption; it is forced by this smoothness. For $h > 0$,

$$\frac{2\varphi_X(0) - \varphi_X(h e_j) - \varphi_X(-h e_j)}{h^2} = \mathbb{E}\left[\frac{2 - 2\cos(hX_j)}{h^2}\right],$$

and the left side converges to $-\partial_j^2 \varphi_X(0) = \Sigma_{jj} + \mu_j^2$ as $h \downarrow 0$. The integrand is non-negative and converges pointwise to X_j^2 , so Fatou gives $\mathbb{E}[X_j^2] \leq \Sigma_{jj} + \mu_j^2 < \infty$.

With $\mathbb{E}|X|^2 < \infty$ the difference quotients of $e^{-iX \cdot \xi}$ are dominated by $|X_j|$ and by $|X_j X_k|$, so we may differentiate under the expectation twice: $\partial_j \varphi_X(0) = -i \mathbb{E}[X_j]$ and $\partial_j \partial_k \varphi_X(0) = -\mathbb{E}[X_j X_k]$. Comparing with the display gives $\mathbb{E}[X_j] = \mu_j$ and $\mathbb{E}[X_j X_k] = \Sigma_{jk} + \mu_j \mu_k$. ■

Theorem 1.5 (Affine images). *If $X \sim \mathcal{N}(\mu, \Sigma)$ and $A \in \mathbb{R}^{m \times d}$, $b \in \mathbb{R}^m$, then*

$$AX + b \sim \mathcal{N}(A\mu + b, A\Sigma A^\top).$$

Proof. $\varphi_{AX+b}(\xi) = e^{-ib \cdot \xi} \mathbb{E}[e^{-iX \cdot (A^\top \xi)}] = e^{-ib \cdot \xi} \varphi_X(A^\top \xi)$. Substituting Definition 1.3 and using $\mu \cdot A^\top \xi = (A\mu) \cdot \xi$ gives $\exp\left(-i(A\mu + b) \cdot \xi - \frac{1}{2}\xi^\top A\Sigma A^\top \xi\right)$. ■

Corollary 1.6 (Rotational invariance). *If $X \sim \mathcal{N}(0, I_d)$ and $O \in O(d)$ then $OX \stackrel{d}{=} X$. More generally, for $X \sim \mathcal{N}(0, \Sigma)$ one has $OX \stackrel{d}{=} X$ precisely when $O\Sigma O^\top = \Sigma$.*

Proof. Theorem 1.5 gives $OX \sim \mathcal{N}(0, O\Sigma O^\top)$, and by Proposition 1.4 that law equals $\mathcal{N}(0, \Sigma)$ exactly when $O\Sigma O^\top = \Sigma$. For $\Sigma = I_d$ this is $OO^\top = I_d$. ■

Invariance is what singles the Gaussian out among product measures — Exercise 1.9 makes that exact. In November, invariance under orthogonal *conjugation* will single out a matrix ensemble in the same way.

Corollary 1.7 (One coordinate at a time). $X \sim \mathcal{N}(\mu, \Sigma)$ if and only if $v \cdot X \sim \mathcal{N}(v \cdot \mu, v^\top \Sigma v)$ for every $v \in \mathbb{R}^d$.

Proof. Forward: take $A = v^\top$ in Theorem 1.5. Conversely, the characteristic function of the scalar $v \cdot X$ evaluated at 1 is $\mathbb{E}[e^{-i v \cdot X}] = \varphi_X(v)$; by hypothesis it equals $\exp(-i v \cdot \mu - \frac{1}{2} v^\top \Sigma v)$, which is Definition 1.3. ■

Exercise 1.8 (TYPE A). Let $X \sim \mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^2$ with $\mu = (1, -1)$ and $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

- Write $\varphi_X(\xi)$ out explicitly and recover $\mathbb{E}[X_1 X_2]$ from it by differentiating, as in Proposition 1.4.
- Use Theorem 1.5 to find the laws of $X_1 + X_2$, of $X_1 - X_2$, and of the pair $(X_1 + X_2, X_1 - X_2)$. Are the two independent?
- Now let Σ be an arbitrary covariance on $\mathbb{R}^2$. For which Σ are $X_1 + X_2$ and $X_1 - X_2$ independent, and for which is there a $v \neq 0$ with $v \cdot X$ almost surely constant?

Exercise 1.9 (TYPE C). Let $d \geq 2$ and let X have independent and identically distributed coordinates with $OX \stackrel{d}{=} X$ for every $O \in O(d)$. Show that $X \sim \mathcal{N}(0, cI_d)$ for some $c \geq 0$. You may use that a characteristic function is continuous.

This is the sentence above, proved: rotational invariance and a product structure together admit nothing but the Gaussian. No moment assumption is needed, and none should be made.

1.3 Existence

Nothing so far exhibits a single Gaussian. One computation supplies all of them.

Lemma 1.10. Let g have density $(2\pi)^{-1/2} e^{-x^2/2}$ on $\mathbb{R}$. Then $\mathbb{E}[e^{-i\lambda g}] = e^{-\lambda^2/2}$; that is, $g \sim \mathcal{N}(0, 1)$.

Proof. Set $f(\lambda) = \mathbb{E}[e^{-i\lambda g}]$. Differentiating under the expectation,

$$f'(\lambda) = \frac{-i}{\sqrt{2\pi}} \int_{\mathbb{R}} x e^{-i\lambda x} e^{-x^2/2} dx.$$

The observation that does the work is $x e^{-x^2/2} = -\frac{d}{dx} e^{-x^2/2}$, which converts the factor of x into a derivative and lets integration by parts move it onto the other factor:

$$f'(\lambda) = \frac{i}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-i\lambda x} \frac{d}{dx} (e^{-x^2/2}) dx = \frac{-i}{\sqrt{2\pi}} \int_{\mathbb{R}} \left(\frac{d}{dx} e^{-i\lambda x} \right) e^{-x^2/2} dx = (-i)(-i\lambda) f(\lambda) = -\lambda f(\lambda).$$

With $f(0) = 1$ this forces $f(\lambda) = e^{-\lambda^2/2}$. ■

Remark 1.11. What the proof used is $\mathbb{E}[g h(g)] = \mathbb{E}[h'(g)]$, with $h(x) = e^{-i\lambda x}$: against a standard Gaussian, multiplication by g acts like differentiation. Remember the move. It returns in Lecture 4 under its own name, and again in Lecture 5, where it proves Wick's theorem.

If $Z = (g_1, \dots, g_d)$ has independent standard normal coordinates then

$$\varphi_Z(\xi) = \prod_{j=1}^d \mathbb{E}[e^{-i\xi_j g_j}] = \prod_{j=1}^d e^{-\xi_j^2/2} = e^{-|\xi|^2/2},$$

so $Z \sim \mathcal{N}(0, I_d)$: the standard Gaussian in d dimensions is a tensor product of one-dimensional ones. Every other Gaussian is an affine image of it.

Proposition 1.12 (Well-posedness). *Let $\mu \in \mathbb{R}^d$ and let Σ be symmetric. A random vector satisfying Definition 1.3 exists if and only if $\Sigma \succeq 0$, and is then unique in law. If $\Sigma = AA^\top$ one may take $X = AZ + \mu$ with $Z \sim \mathcal{N}(0, I_d)$.*

Proof. If $\Sigma \succeq 0$, factor $\Sigma = AA^\top$. Theorem 1.5 applied to $Z \sim \mathcal{N}(0, I_d)$ gives $AZ + \mu \sim \mathcal{N}(\mu, AA^\top) = \mathcal{N}(\mu, \Sigma)$. Uniqueness in law is Proposition 1.1(iv).

Conversely, suppose $v^\top \Sigma v = -c$ with $c > 0$. Along $\xi = sv$ the right side of Definition 1.3 has modulus $e^{s^2 c/2} \rightarrow \infty$, contradicting $|\varphi_X| \leq 1$. ■

The family of Gaussian laws on $\mathbb{R}^d$ is therefore indexed exactly by $\mathbb{R}^d \times \text{Sym}_{\geq 0}(\mathbb{R}^d)$, and the representation $X = AZ + \mu$ is a construction inside a proof rather than a rival definition.

Example 1.13 (The degenerate case). Take $d = 2$, $\mu = 0$ and $\Sigma = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, positive semidefinite of rank one. By Proposition 1.12 the law $\mathcal{N}(0, \Sigma)$ exists, and $A = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ realizes it as $X = (g, g)$. It has no density — it is supported on a line — and Definition 1.3 never notices. A definition through the density $\propto e^{-\frac{1}{2}x^\top \Sigma^{-1}x}$ would have had to exclude it, and there is nothing pathological about it to exclude.

Exercise 1.14 (TYPE A). Let $f(x) = (2\pi)^{-d/2} e^{-|x|^2/2}$ on $\mathbb{R}^d$. Show $\widehat{f}(\xi) = e^{-|\xi|^2/2}$, then verify Proposition 1.1(i) on this pair by evaluating $(2\pi)^{-d} \int_{\mathbb{R}^d} e^{-|\xi|^2/2} e^{ix \cdot \xi} d\xi$ and recovering f . Conclude that the standard Gaussian density is a fixed point of the Fourier transform up to the factor $(2\pi)^{d/2}$.

Exercise 1.15 (TYPE B). Let $\Sigma \succ 0$, write $\Sigma = AA^\top$ with A invertible, and let $X = AZ + \mu$.

- Change variables in the density of Z to obtain $p_X(x) = (2\pi)^{-d/2} (\det \Sigma)^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right)$, and check that the answer does not depend on which square root A was chosen.
- Show that when Σ is singular no density exists: produce $v \neq 0$ with $v \cdot (X - \mu) = 0$ almost surely, and conclude that X lives on an affine subspace of dimension less than d .

Part (a) is the definition most students meet first. Part (b) is what it cannot say, and Example 1.13 is the smallest instance.

1.4 Uncorrelated versus independent

For general random variables zero covariance says very little. For jointly Gaussian ones it says everything.

Theorem 1.16. *Let (X, Y) be jointly Gaussian in $\mathbb{R}^d \times \mathbb{R}^{d'}$ — the concatenated vector is Gaussian — with $\text{Cov}(X, Y) = 0$. Then $X \perp\!\!\!\perp Y$.*

Proof. Write $\xi = (s, u)$. The covariance of the concatenated vector is block diagonal, so $\xi^\top \Sigma \xi = s^\top \Sigma_{XX} s + u^\top \Sigma_{YY} u$ and $\mu \cdot \xi = \mu_X \cdot s + \mu_Y \cdot u$. Hence

$$\varphi_{(X,Y)}(s, u) = \varphi_X(s) \varphi_Y(u),$$

which is the characteristic function of the product law $\mathcal{L}(X) \otimes \mathcal{L}(Y)$. By Proposition 1.1(iv) the joint law *is* that product. ■

Uniqueness has now done two different jobs in one lecture. The hypothesis that fails in practice is joint Gaussianity, not Gaussianity of the marginals.

Example 1.17 (Uncorrelated, standard normal, not independent). Let $g \sim \mathcal{N}(0, 1)$ and let ε be an independent random sign; set $X = g$ and $Y = \varepsilon g$. Conditionally on $\varepsilon = \pm 1$ we have $Y = \pm g \stackrel{d}{=} g$, so both marginals are standard normal. They are uncorrelated, since $\mathbb{E}[XY] = \mathbb{E}[\varepsilon]\mathbb{E}[g^2] = 0$. And $|X| = |Y|$ always, so X determines Y up to a sign.

Conditioning on ε and applying Lemma 1.10,

$$\varphi_{(X,Y)}(s, u) = \frac{1}{2}e^{-(s+u)^2/2} + \frac{1}{2}e^{-(s-u)^2/2},$$

a sum of two Gaussian transforms and not $\exp(\text{quadratic})$. The pair fails Definition 1.3 even though each coordinate satisfies it.

Exercise 1.18 (TYPE B). Continue Example 1.17.

- (a) Find the law of $X + Y$ and show it is not Gaussian, by exhibiting an atom.
- (b) Corollary 1.7 says a vector is Gaussian as soon as every linear functional of it is. Compute the law of $sX + uY$ for arbitrary s, u and say why there is no contradiction.
- (c) Does (X, Y) have a density on $\mathbb{R}^2$?

1.5 * Entropy, and why the Gaussian maximizes it

Not lectured. Exercises due at the start of Lecture 2.

Definition 1.3 is not the only characterization of the Gaussian. Among all laws on $\mathbb{R}^d$ with a given mean and covariance $\Sigma \succ 0$, it is the one of largest differential entropy — the least committal law consistent with a second moment. That is what this section proves, and Lecture 3 is where it gets used: it is the answer to *why a Gaussian at all*.

Before entropy, surprise. If an event of probability p occurs, the surprise $s(p)$ ought to satisfy $s(1) = 0$ and $s(pq) = s(p) + s(q)$. Substituting $p = e^{-u}$ turns the second into $g(u+v) = g(u) + g(v)$ for $g(u) = s(e^{-u})$, whose only continuous solutions are linear. So $s(p) = -c \log p$, the base of the logarithm is a choice of units, and we take $c = 1$.

The **entropy** of a law on a finite set is the surprise expected before looking,

$$H(p) = \mathbb{E}[-\log p(X)] = -\sum_x p(x) \log p(x),$$

and for a density p on $\mathbb{R}^d$ the same formula defines the **differential entropy** $h(p) = -\int p \log p$. The second is not the first in disguise, and the next exercise is where that becomes visible.

Exercise 1.19 (TYPE A). (a) Compute H for a coin of bias q ; sketch it on $q \in [0, 1]$ and find where it is largest. Compute H for the uniform law on n points.

- (b) Compute h for the uniform law on $[0, a]$, the exponential of rate λ , and $\mathcal{N}(0, \sigma^2)$ — you should find $\frac{1}{2} \log(2\pi e \sigma^2)$ for the last. Exhibit a law with *negative* differential entropy, and say why no discrete entropy can be negative.
- (c) Show $h(aX) = h(X) + \log |a|$ for $a \neq 0$. So “the entropy of a length” is not a well-defined number, while H has no units at all.

The pattern we are after is this: *fix a constraint, and entropy is maximized by the least committal law satisfying it*. With no constraint on a finite set that is the uniform law. The Gaussian will answer the same question with a different constraint — but part (c) shows h is the wrong quantity

to state the answer with, and the cure is to measure entropy *relative* to a reference. For densities p, q , the **relative entropy** is $D(p \parallel q) = \int p \log(p/q)$: the extra surprise from expecting q when the truth is p , so it should never be negative.

Exercise 1.20 (TYPE B). Prove **Gibbs' inequality**: $D(p \parallel q) \geq 0$, with equality iff $p = q$. *Jensen applied to $-\log$* . Deduce the discrete instance $H(p) \leq \log n$ for every law on n points, with equality only for the uniform one, by taking q uniform. Then check that $D(p \parallel q)$ is unchanged when X is rescaled — unlike h .

Now fix μ and $\Sigma \succ 0$, let q be the Gaussian density, and let p be any density with that mean and covariance. Expanding $0 \leq D(p \parallel q)$ gives $0 \leq -h(p) - \int p \log q$, and everything hinges on the second term.

Exercise 1.21 (TYPE B). Write out $\log q$ and observe it is a *quadratic polynomial* in x . Deduce that $\int p \log q$ depends on p only through its mean and covariance, hence equals $\int q \log q = -h(q)$. Conclude $h(p) \leq h(q)$ with equality only when $p = q$, and check $h(q) = \frac{1}{2} \log((2\pi e)^d \det \Sigma)$.

Note how little the Gaussian had to do: everything turned on $\log q$ being quadratic, so that integrating it against p could see only p 's first two moments. *A quadratic is exactly what a covariance constraint measures.*

Exercise 1.22 (TYPE C). Let X, Y be i.i.d., *symmetric*, each with a density. Show

$$h\left(\frac{X+Y}{\sqrt{2}}\right) \geq h(X).$$

You may use subadditivity $h(U, V) \leq h(U) + h(V)$, which is Gibbs' inequality applied to a joint density against the product of its marginals. Lecture 3 opens with this inequality.

Exercise 1.23 (TYPE C). Fix $S \subseteq \mathbb{R}^d$, functions $T_1, \dots, T_k$ on S and numbers $c_1, \dots, c_k$. **Assume** that for some $\lambda_1, \dots, \lambda_k$ the function $\exp(\sum_m \lambda_m T_m)$ has finite integral Z over S , so that $q = Z^{-1} \exp(\sum_m \lambda_m T_m)$ is a density, and that $\int q T_m = c_m$ for each m . Show q uniquely maximizes h among densities on S with those moments.

Then reread the paragraph above — *which property of the Gaussian did that argument use?* Recover as instances the uniform law on $[0, a]$, the exponential at fixed mean on $[0, \infty)$, and $\mathcal{N}(\mu, \Sigma)$ at fixed mean and covariance. Finally, show the assumption does real work: on $S = \mathbb{R}$ with the mean alone constrained no λ makes $e^{\lambda x}$ integrable, and there is no maximizer — exhibit densities of mean 0 with h arbitrarily large.

2 Conditioning as projection

Q. We observe Y . What is our best guess for X ?

2.1 Expectation is a projection

Let $Z \sim \nu$ be a random vector in $\mathbb{R}^k$ and let $f : \mathbb{R}^k \rightarrow \mathbb{R}$ satisfy $\mathbb{E}[f(Z)^2] < \infty$.

Lemma 2.1. $\mathbb{E}[f(Z)] = \arg \min_{c \in \mathbb{R}} \mathbb{E}[(f(Z) - c)^2]$, and the minimum value is $\text{Var}(f(Z))$.

Proof. $\mathbb{E}[(f(Z) - c)^2] = \mathbb{E}[(f(Z) - \mathbb{E}f(Z))^2] + (\mathbb{E}f(Z) - c)^2$, since the cross term is $2(\mathbb{E}f(Z) - c) \mathbb{E}[f(Z) - \mathbb{E}f(Z)] = 0$. ■

Two readings of the same statement.

First: $\mathbb{E}[f(Z)]$ is the best deterministic guess for $f(Z)$ under squared error.

Second: write

$$L^2(\nu) = \left\{ f : \mathbb{R}^k \rightarrow \mathbb{R} \mid \mathbb{E}[f(Z)^2] < \infty \right\}, \quad \langle f, g \rangle_\nu = \mathbb{E}[f(Z)g(Z)],$$

with functions agreeing ν -almost everywhere identified, so that $\langle \cdot, \cdot \rangle_\nu$ is an inner product. Then $\mathbb{E}[(f(Z) - c)^2] = \|f - c\|_\nu^2$, and Lemma 2.1 says

$$\mathbb{E}[f(Z)] = \text{orthogonal projection of } f \text{ onto the constants } \subseteq L^2(\nu).$$

Imported: the projection theorem. $L^2(\nu)$ is complete, and for a closed subspace $H \subseteq L^2(\nu)$ every f has a unique decomposition $f = \Pi_H f + r$ with $\Pi_H f \in H$ and $r \perp H$. Equivalently, $\Pi_H f$ is the unique minimizer of $\|f - h\|_\nu$ over $h \in H$.

2.2 Conditional expectation

Now let $Z = (X, Y) \sim \nu$ with $X \in \mathbb{R}$ and $Y \in \mathbb{R}^m$, and write ν_Y for the law of Y . A function $g \in L^2(\nu_Y)$ becomes a function of Z through $(x, y) \mapsto g(y)$, and

$$\mathbb{E}[g(Y)^2] = \int g^2 d\nu_Y$$

is the same number computed either way. So this map is a linear isometry of $L^2(\nu_Y)$ onto a subspace of $L^2(\nu)$, and that subspace is complete, hence closed. We identify $L^2(\nu_Y)$ with its image.

Definition 2.2 (Conditional expectation). $\mathbb{E}[X \mid Y]$ is the orthogonal projection of the coordinate function $(x, y) \mapsto x$ onto $L^2(\nu_Y)$.

It is by construction an element of $L^2(\nu_Y)$: a function of Y , determined up to ν_Y -null sets. The projection theorem gives two equivalent descriptions.

Lemma 2.3. For $m \in L^2(\nu_Y)$ the following are equivalent:

- (i) $m = \mathbb{E}[X \mid Y]$;
- (ii) $\mathbb{E}[(X - m(Y))g(Y)] = 0$ for every $g \in L^2(\nu_Y)$;
- (iii) m minimizes $\mathbb{E}[(X - g(Y))^2]$ over $g \in L^2(\nu_Y)$.

Moreover, for every $g \in L^2(\nu_Y)$,

$$\mathbb{E}[(X - g(Y))^2] = \mathbb{E}[(X - \mathbb{E}[X \mid Y])^2] + \mathbb{E}[(\mathbb{E}[X \mid Y] - g(Y))^2]. \quad (1)$$

Proof. The equivalence and the identity are the projection theorem and Pythagoras. ■

For $X \in \mathbb{R}^p$ apply the definition to each coordinate.

2.3 The tower property

Proposition 2.4. *Let $Z = (X, Y, V)$. Then*

$$\mathbb{E}[\mathbb{E}[X | Y, V] | Y] = \mathbb{E}[X | Y], \quad \mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X],$$

and $\mathbb{E}[g(Y)X | Y] = g(Y)\mathbb{E}[X | Y]$ for bounded g .

Proof. $L^2(\nu_Y) \subseteq L^2(\nu_{(Y,V)})$, both closed in $L^2(\nu)$. For nested closed subspaces $H_1 \subseteq H_2$ and any f , the vector $\Pi_{H_2}f - f$ is orthogonal to H_2 and hence to H_1 , so $\Pi_{H_1}\Pi_{H_2}f = \Pi_{H_1}f$. The first identity is this with $H_1 = L^2(\nu_Y)$, $H_2 = L^2(\nu_{(Y,V)})$; the second is this with H_1 the constants, $H_2 = L^2(\nu_Y)$. For the last, $\mathbb{E}[(g(Y)X - g(Y)\mathbb{E}[X | Y])h(Y)] = \mathbb{E}[(X - \mathbb{E}[X | Y])(gh)(Y)] = 0$ for every h . ■

Corollary 2.5 (Law of total variance). *With $\text{Var}(X | Y) := \mathbb{E}[(X - \mathbb{E}[X | Y])^2 | Y]$,*

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y]).$$

Proof. Take $g \equiv \mathbb{E}X$ in (1). ■

Exercise 2.6 (TYPE A). Show $\text{Var}(\mathbb{E}[X | Y]) \leq \text{Var}(X)$. Characterize equality, and characterize the case in which the first term of Corollary 2.5 carries all of the variance.

Exercise 2.7 (TYPE B). Let J be a fair coin and, given J , let $X \sim \mathcal{N}(\pm 1, \sigma^2)$ accordingly. Compute both terms of Corollary 2.5 with $Y = J$ and check the total against $\text{Var}(X)$. Then let W be a function of $X_1, \dots, X_n$ and suppose $\mathbb{E}[U | X_1, \dots, X_n] = -W/n$. Deduce $\mathbb{E}[U | W] = -W/n$, naming the property used.

2.4 Gaussian conditioning

Let $(X, Y) \sim \mathcal{N}(0, \Sigma)$ in $\mathbb{R}^p \times \mathbb{R}^m$, with blocks $\Sigma_{XX}, \Sigma_{XY} = \mathbb{E}[XY^\top], \Sigma_{YY} = \mathbb{E}[YY^\top]$, so that

$$\mathbb{E}[e^{-i(\xi \cdot X + \eta \cdot Y)}] = \exp\left(-\frac{1}{2}\xi^\top \Sigma_{XX}\xi - \xi^\top \Sigma_{XY}\eta - \frac{1}{2}\eta^\top \Sigma_{YY}\eta\right), \quad \xi \in \mathbb{R}^p, \eta \in \mathbb{R}^m. \quad (2)$$

Lemma 2.8. *There exists $A \in \mathbb{R}^{p \times m}$ with $A\Sigma_{YY} = \Sigma_{XY}$; any two solutions satisfy $A_1Y = A_2Y$ almost surely; and if $\Sigma_{YY} \succ 0$ the solution is unique and equals $\Sigma_{XY}\Sigma_{YY}^{-1}$.*

Proof. Transposed, the equation is $\Sigma_{YY}A^\top = \Sigma_{YX}$, solvable exactly when each column $\mathbb{E}[YX_i]$ of Σ_{YX} lies in range $\Sigma_{YY} = (\ker \Sigma_{YY})^\perp$. If $v \in \ker \Sigma_{YY}$ then $\text{Var}(v \cdot Y) = v^\top \Sigma_{YY}v = 0$, so $v \cdot Y = 0$ almost surely and $v^\top \mathbb{E}[YX_i] = \mathbb{E}[(v \cdot Y)X_i] = 0$. If $B = A_1 - A_2$ then $\text{Cov}(BY) = B\Sigma_{YY}B^\top = 0$, so $BY = 0$ almost surely. ■

Theorem 2.9. *With A as in Lemma 2.8, $\mathbb{E}[X | Y] = AY$.*

Proof. $\mathbb{E}[X | Y]$ is a function of Y , and a function of Y is determined ν_Y -almost everywhere by the numbers $\mathbb{E}[e^{-i\eta \cdot Y}h(Y)]$, $\eta \in \mathbb{R}^m$, by Proposition 1.1(iv). So it suffices to compute them. Lemma 2.3(ii) with $g(y) = e^{-i\eta \cdot y}$ removes the conditioning, and (2) does the rest:

$$\mathbb{E}[e^{-i\eta \cdot Y} \mathbb{E}[X | Y]] = \mathbb{E}[e^{-i\eta \cdot Y} X] = i \partial_\xi|_{\xi=0} \mathbb{E}[e^{-i\xi \cdot X - i\eta \cdot Y}] = -i \Sigma_{XY} \eta \varphi_Y(\eta).$$

Now run the same three steps with AY in place of X . The pair (AY, Y) is Gaussian with $\mathbb{E}[(AY)Y^\top] = A\Sigma_{YY}$, so

$$\mathbb{E}[e^{-i\eta \cdot Y} AY] = -i A\Sigma_{YY} \eta \varphi_Y(\eta).$$

The two agree for every η exactly when $A\Sigma_{YY} = \Sigma_{XY}$. ■

Not differentiating gives the whole conditional law by the same computation.

Theorem 2.10. Put $\Sigma_{X|Y} = \Sigma_{XX} - A\Sigma_{YX}$. Then for every $\xi \in \mathbb{R}^p$, almost surely,

$$\mathbb{E}[e^{-i\xi \cdot X} | Y] = \exp\left(-i\xi \cdot AY - \frac{1}{2}\xi^\top \Sigma_{X|Y} \xi\right),$$

that is, $X | Y \sim \mathcal{N}(AY, \Sigma_{X|Y})$. In particular $\text{Cov}(X | Y) = \Sigma_{X|Y}$ does not depend on Y , and $X - AY$ is independent of Y .

Proof. Complete the square in η in (2). Since $A\Sigma_{YY} = \Sigma_{XY}$,

$$\frac{1}{2}\eta^\top \Sigma_{YY} \eta + \xi^\top \Sigma_{XY} \eta = \frac{1}{2}(\eta + A^\top \xi)^\top \Sigma_{YY} (\eta + A^\top \xi) - \frac{1}{2}\xi^\top A\Sigma_{YY} A^\top \xi,$$

and $A\Sigma_{YY} A^\top = A\Sigma_{YX}$, so $\Sigma_{XX} - A\Sigma_{YX} A^\top = \Sigma_{X|Y}$. Using $(A^\top \xi) \cdot Y = \xi \cdot AY$,

$$\mathbb{E}[e^{-i\eta \cdot Y} e^{-i\xi \cdot X}] = e^{-\frac{1}{2}\xi^\top \Sigma_{X|Y} \xi} \mathbb{E}[e^{-i(\eta + A^\top \xi) \cdot Y}] = \mathbb{E}[e^{-i\eta \cdot Y} e^{-i\xi \cdot AY - \frac{1}{2}\xi^\top \Sigma_{X|Y} \xi}]$$

for every η , and the argument of Theorem 2.9 identifies the two functions of Y . The right side depends on Y only through the factor $e^{-i\xi \cdot AY}$, so $\mathbb{E}[e^{-i\xi \cdot (X - AY)} | Y]$ is constant; multiplying by $e^{-i\eta \cdot Y}$ and taking expectations factors the joint transform of $(X - AY, Y)$. ■

When $\Sigma_{YY} \succ 0$,

$$AY = \Sigma_{XY} \Sigma_{YY}^{-1} Y, \quad \Sigma_{X|Y} = \Sigma_{XX} - \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX},$$

the *Schur complement* of Σ_{YY} in Σ . For uncentered (X, Y) replace AY by $\mu_X + A(Y - \mu_Y)$. Nowhere is the null event $\{Y = y\}$ conditioned on.

2.5 Bayesian linear models and ridge regression

Take

$$y = Hw + \sigma\varepsilon, \quad w \sim \mathcal{N}(0, \tau^2 I_p), \quad \varepsilon \sim \mathcal{N}(0, I_n) \text{ independent of } w.$$

The pair (w, y) is a linear image of (w, ε) , hence jointly Gaussian by Theorem 1.5, with

$$\Sigma_{ww} = \tau^2 I, \quad \Sigma_{wy} = \tau^2 H^\top, \quad \Sigma_{yy} = \tau^2 H H^\top + \sigma^2 I.$$

Theorem 2.10 gives the posterior; simplifying with $\tau^2 H^\top (\tau^2 H H^\top + \sigma^2 I)^{-1} = (H^\top H + \lambda I)^{-1} H^\top$:

Proposition 2.11. With $\lambda = \sigma^2/\tau^2$,

$$\mathbb{E}[w | y] = (H^\top H + \lambda I)^{-1} H^\top y, \quad \text{Cov}(w | y) = \sigma^2 (H^\top H + \lambda I)^{-1}.$$

The posterior mean is ridge regression, with regularization parameter the ratio of noise variance to prior variance.

The exercises below are due at the start of Lecture 3.

Exercise 2.12 (TYPE A). (a) Show $-\log p(y | w) = \frac{1}{2\sigma^2} \|y - Hw\|^2 + c$ with c free of w , so least squares is maximum likelihood. State the assumption on the noise that choosing squared error encodes.

(b) Show the maximizer of the posterior density is $\hat{w}_\lambda = \arg \min_w \|y - Hw\|^2 + \lambda \|w\|^2$. Interpret large λ , and compute the limits $\tau \rightarrow 0$ and $\tau \rightarrow \infty$.

Exercise 2.13 (TYPE B). Compute the three blocks of the joint covariance of (w, y) and apply Theorem 2.10 to obtain Proposition 2.11. Then verify that $\text{Cov}(w | y)$ does not depend on y , and say what that means for an experimenter who observes a surprising y .

Exercise 2.14 (TYPE B). Let $Y \sim \mathcal{N}(0, 1)$ and $X = Y^2 - 1$. Show $\text{Cov}(X, Y) = 0$, so the projection of X onto $\text{span}\{1, Y\}$ is 0, with mean squared error 2. Compute $\mathbb{E}[X | Y]$ and its error. Which hypothesis of Theorem 2.9 fails?

2.6 * The two forms of the ridge estimator

Not lectured. Proposition 2.11 inverts a $p \times p$ matrix; the derivation that produced it inverted an $n \times n$ one.

Exercise 2.15 (TYPE C). Show that for $\lambda > 0$

$$(H^\top H + \lambda I_p)^{-1} H^\top = H^\top (H H^\top + \lambda I_n)^{-1},$$

and deduce that ridge regression with $p \gg n$ costs $O(n^3)$ rather than $O(p^3)$. Then take $\tau \rightarrow \infty$ with $n < p$ and H of full row rank, and show the posterior mean converges to $H^\top (H H^\top)^{-1} y$, the minimum-norm solution of $Hw = y$. Explain why the limit is visible in one of the two forms and not the other.

2.7 * Denoising is conditioning

Not lectured. Let X be a random vector in $\mathbb{R}^d$ with $\mathbb{E}|X|^2 < \infty$, not assumed Gaussian, and

$$Y = X + \sigma \xi, \quad \xi \sim \mathcal{N}(0, I_d) \text{ independent of } X.$$

Then Y has density $p_Y(y) = \mathbb{E}[\phi_\sigma(y - X)]$, where ϕ_σ is the $\mathcal{N}(0, \sigma^2 I_d)$ density. The function $\nabla \log p_Y$ is the *score* of Y .

Proposition 2.16 (Tweedie's formula).

$$\mathbb{E}[X | Y] = Y + \sigma^2 \nabla \log p_Y(Y).$$

Proof. By Lemma 2.3(ii) it suffices to show $\mathbb{E}[\langle X - Y - \sigma^2 \nabla \log p_Y(Y), f(Y) \rangle] = 0$ for every smooth compactly supported f .

Conditionally on X the vector ξ is standard Gaussian, so $\mathbb{E}[\eta F(\eta)] = \mathbb{E}[F'(\eta)]$ — Lemma 1.10, in each coordinate — gives

$$\mathbb{E}[\langle Y - X, f(Y) \rangle] = \sigma \mathbb{E}[\langle \xi, f(X + \sigma \xi) \rangle] = \sigma^2 \mathbb{E}[\nabla \cdot f(Y)].$$

Integrating by parts in y ,

$$\mathbb{E}[\nabla \cdot f(Y)] = \int (\nabla \cdot f) p_Y dy = - \int \langle f(y), \nabla p_Y(y) \rangle dy = - \mathbb{E}[\langle f(Y), \nabla \log p_Y(Y) \rangle]. \quad \blacksquare$$

Exercise 2.17 (TYPE A). Take $X \sim \mathcal{N}(0, \tau^2 I_d)$. Compute p_Y and $\nabla \log p_Y$, and check that Proposition 2.16 gives $\mathbb{E}[X | Y] = \frac{\tau^2}{\tau^2 + \sigma^2} Y$, in agreement with Theorem 2.9. Which of the two derivations used that X is Gaussian?

Exercise 2.18 (TYPE B). Show p_Y is smooth with $\nabla p_Y(y) = \mathbb{E}[\nabla \phi_\sigma(y - X)]$, and state what is required for the boundary term in the integration by parts to vanish. Then show that testing against smooth compactly supported f suffices in Lemma 2.3(ii).

Exercise 2.19 (TYPE C). A *denoiser* is a minimizer D of $\mathbb{E}|X - D(Y)|^2$ over all functions $D : \mathbb{R}^d \rightarrow \mathbb{R}^d$. Show that D and $\nabla \log p_Y$ determine each other explicitly, and deduce that fitting a denoiser by least squares to pairs $(X, X + \sigma\xi)$ — which requires neither p_X nor a likelihood — estimates the score of the law of Y .